

Appearance Manifold User Guide

Using Weathering Manager Component to configure and apply real-time weathering textures

Audience

This guide is for artists, level designers, and Blueprint developers. It focuses on component setup, keyframe preparation, and runtime validation instead of the internal C++ implementation.

1. Overview

Appearance Manifold converts BaseColor, Specular, and Roughness textures into a 7D feature representation, then generates intermediate runtime weathering states along a precomputed Weathering Trajectory.

Users attach Weathering Manager Component to an actor, prepare keyframe textures and a trajectory, then control runtime progression with Weathering Speed and Step values.

2. Requirements

Item	Description
Target actor	An actor with a Static Mesh or MeshComponent that will receive the weathered material result.
Sample textures	Sample BaseColor, Sample Specular, and Sample Roughness. All three textures must have the same resolution.
Material	A material with Base/Spec/Rough texture parameters. Recommended default names are Weathering_Base, Weathering_Spec, and Weathering_Rough.
Recommended resolution	512 x 512. This is the most stable baseline for quality and runtime cost in the current test workflow.
Trajectory	A runtime weathering path binary generated by Generate Weathering Trajectory.

3. Basic Workflow

- 1 Add Weathering Manager Component to the actor that should receive weathering.
- 2 Assign MeshComponent and MaterialIndex.
- 3 Assign Sample BaseColor, Sample Specular, Sample Roughness, and Sample Size.
- 4 Match BaseParamName, SpecParamName, and RoughParamName with the actual material parameter names.
- 5 Run Clear Key Frame Texture to reset existing keyframe cache data.

- 6 Run Get New Key Frame to store the current texture state as a keyframe.
- 7 Change the weathered texture state, for example through GammaTon simulation, then run Get New Key Frame again.
- 8 Run Print Key Frame Count to confirm that keyframes were stored.
- 9 Run Generate Weathering Trajectory to generate the runtime interpolation path.
- 10 Enter Play mode, enable Allow Weathering, then tune Weathering Speed and Max Weathering Step.

4. Main Component Options

Option	Purpose
Allow Weathering	Enables or disables runtime weathering progression.
Weathering Speed	Controls how quickly the weathering step advances.
Weathering Step	Current weathering progression step. It is used as the basis for alpha calculation.
Max Weathering Step	Maximum number of progression steps. Higher values allow finer progression.
Cached Nearest A/B	Cached nearest trajectory indices for keyframe pixels. They should be ready before interpolation starts.
Out Base/Spec/Rough	Transient runtime textures reconstructed from the interpolated 7D tensor.

5. Runtime Flow

The component repeats the following process through Tick or user-driven events.

- Check whether Allow Weathering is enabled.
- Update Weathering Step and calculate Alpha.
- Generate an intermediate 7D tensor with InterpolateWeatheringCached.
- Reconstruct BaseColor, Specular, and Roughness textures with ReconstructTexturesFromTensor.
- Apply the textures to a Dynamic Material Instance with ApplyWeatheringTexturesToMesh.

Recommended pattern

Generate keyframes and trajectory in the editor first. At runtime, use only cached trajectory and tensor data for interpolation.

6. Troubleshooting

Symptom	What to check
Textures not visible	Check MeshComponent, MaterialIndex, and material parameter names.
Distorted result	Confirm that BaseColor, Specular, Roughness, and keyframe textures share the same resolution.
Frame hitching	Lower texture resolution or reduce how often Weathering Step updates.
Trajectory generation fails	Check PythonScriptPlugin, numpy, scipy, scikit-learn, and the Unreal Output Log.
Empty interpolation result	Confirm that TrajectorySamples, CachedNearestA, and CachedNearestB are prepared.

7. Quick Checklist

- All three textures have the same resolution.
- Sample Size intentionally matches the texture size.
- Keyframe count was confirmed with Print Key Frame Count.
- Generate Weathering Trajectory completed successfully.
- Material parameter names match the actual material.
- Allow Weathering is enabled and Weathering Speed is not zero.